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COMPUTER ESTIMATED ALPHA ATTENUATION DURING PROBLEM SOLVING IN CHILDREN WITH LEARNING DISABILITIES

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Children with learning disabilities (LD) have been consistently described as distractible and showing marked attention deficits (Clements 1966; Dykman et al. 1971; Hallahan and Cruickshank 1973). There is currently research evidence to support the view that defective attention is central to learning disabilities (Luria 1961; Dykman et al. 1970, 1971; Anderson et al. 1973).

The alpha rhythm originally described by Berger (1929, 1931) was recognized by him to reduce in amplitude or obliterate (now commonly referred to as "attenuation" or "blocking", respectively) with the onset of alert attention. This well-known effect has been regularly described (Adrian and Matthews 1934; Mulholland 1969; Kooi 1971) and has been incorporated into the official definition of the "alpha rhythm" of the "Glossary prepared for the International Federation for Electroencephalography and Clinical Neurophysiology" (Chatrian et al. 1974). Rheinberger and Jasper (1937) also observed that this effect slowly decreased as attention toward the stimulus waned. Thus, if there are attention deficits present in children with learning disabilities, there may be accompanying changes in ability to attenuate to alpha rhythms.

The present study was undertaken to determine whether or not LD children show less alpha attenuation when engaged in mental arithmetic, immediate memory tasks, compared to a matched group of normal control boys.

Method and material

The subjects were 10 boys with learning disabilities, 10.1—12.6 years old (M = 11.3), and 11 normal control boys, 10.6—11.7 years old (M = 11.2), from the public school system. The experimental subjects were from special LD classes and the controls were from regular classrooms. A lower age limit of 10 years was decided on because by the tenth year the occipital alpha band is fairly well established at a mean value of 9.5 c/sec, and a range of 8—10 c/sec (Brazier 1968; Kooi 1971).

The experimental boys were selected from the broad controversial and currently loosely defined category of Learning Disability (LD) and Minimal Brain Dysfunction (MBD) (REF), "SLD/MDB". Because of the current confusion and disagreement in terminology, the following eligibility criteria were used to define operationally the term learning disability: (1) A WISC Full Scale I.Q. of 90 or above. (2) Two grade levels low in reading and one grade level low in arithmetic on the Wide Range Achievement Test (WRAT). (3) The presence of attention deficits, distractibility, hyperactivity, or other signs of marked inability to attend, as rated by the teacher and/or specialist on a 5-point rating scale (1 = minimal to 5 = extensive). A rating of "4" or "5" constituted eligibility on this item. (4) No specific visual, hearing or motor handicaps. (5) No history of epilepsy. (6) No brain injury.

The control subjects were matched on all the above items except item "2", for which

they were at expected levels for age and grade on the Metropolitan Achievement Test (MAT), and item "3" for which there was no reported behavioral evidence. Adherence to the Federal Family Education Rites and Privacy Act, and local district policy did not allow the investigators access to individual scores to establish means and standard deviations.

All EEG recordings were made in a shielded room with the child sitting in a chair. Recordings were made on an 8-channel electroencephalograph. Disc electrodes filled with conducting jelly were placed on the scalp according to the International 10—20 system at P3, P4, O1, O2, and an electrode near CZ served as ground. The above electrode placements constituted two bipolar parieto-occipital derivations for the left and right hemispheres respectively.

The EEGs were recorded at a sensitivity of 7.5 μV/mm, with a frequency band of 0.5—50 c/sec (—3 dB), at a paper speed of 30 mm/sec. The EEG was also recorded on an FM tape recorder.

The experimenter explained to each child how the recording was done and each step in preparation, and the child was encouraged to talk about his expectations to minimize anxiety. Each child was also told that he would receive $ 2.00 for participating and doing the tasks, but that he could decide not to participate or stop at any time. After being seated and having the electrodes placed, he received instructions to sit quietly with eyes closed while a 5 min baseline recording was made. The child was also requested twice towards the end of this recording to open his eyes and look directly ahead at a designated point on the wall for a few sec and then again close the eyes. This test was to determine the degree of each subject's alpha attenuating ability. Instructions for performing the tasks and verbal reporting were then given, and cotton balls were taped over the eyes. The programmed taped instructions and problems were begun. Half of the subjects got instruction tape "A" and the other half tape "B", with the order of the computational and reasoning requirements in reverse on the two tapes in order to control for any order effect.

Problems were chosen which involved reasoning and pure addition arithmetic to insure that the tasks were relevant to language skills as well as to computations. An example of a problem presented is the following: "If James has 25 cents and spends 15 cents on penny candy, how many cents does he have left?" The recall tasks each consisted of five items of a like kind to be repeated back as heard.

Upon completion, recordings were checked and those that did not have the following characteristics were eliminated from the study: (1) a resting average alpha frequency between 8 and 13 c/sec; (2) attenuation when the eyes were open as opposed to closed during the baseline recording. Alpha attenuation was scored when there was a manually measured amplitude decrease of at least 75% when the eyes were opened for at least 0.2 sec, originally termed "blocking" by Mulholland and Peper (1971) and Mulholland (1972). Ten out of 13 LD children and 11 out of 14 control subjects satisfactorily met the above criteria, so these were retained in the study.

The FM magnetic tape was then digitized at 200 samples/sec. The digitized data were analyzed by computer using the Radix 2 Fast Fourier Transform Program RFT 1 (Cooley and Tukey 1965). The EEGs from channels 1 and 2 were transformed in blocks of 256 samples (1.28 sec) to produce power spectra which were band averaged separately for the theta (4 to under 8 c/sec), alpha (8—13 c/sec), and beta (over 13 c/sec) (Chatrian et al. 1974) activity for each channel. Channel 3 was a marker channel. All EEG data recorded during problem working were analyzed and all non-problem EEG data were deleted. The computer output was checked for artifacts using the direct written polygraph records and disregarded in the final average, if present.

The average (mean) power was calculated for the alpha band of the left hemisphere during immediate recall (T_1), mental arithmetic (T_2), the initial taped instruction period (T_3),

and the baseline resting interval. The left side was used because language and arithmetic tasks have been shown to engage primarily the left hemisphere (Galin and Ornstein 1972; Doyle et al. 1974).

To control for an observable large interindividual variability in EEG average power in the subjects, the ratio of average power during the various tasks to the power during the resting condition was calculated, allowing power to be described as a ratio with an absolute zero point. The ratios were transformed in order better to meet the assumption of multivariate normality (Morrison 1967), by taking the natural logarithm of the above ratios (see Table II). The Hotelling T^2 test was used to determine whether or not the two groups arose from populations with common means.

Results

Predicted directional differences in means between LD and control children were uniform over the three conditions. Table I shows data for each individual, as well as the group means. Eight of the 10 experimental LD boys failed to show alpha attenuation in one or more of the three conditions, while 9 of the 11 normal control boys showed alpha attenuation under all three conditions. Table II shows the mean score for LD and normal control boys in each condition. Negative scores in this table signify alpha attenuation during the task condition. The control group, but not the LD group, showed alpha attenuation in the T_1 and T_2 conditions, and the control group showed greater alpha attenuation in the

TABLE I

Ratios of average EEG power during each of three tasks (T_1, immediate recall; T_2, mental arithmetic; T_3, initial instructions) to the average EEG power of the rest period for the experimental LD children and normal controls.

Subjects		Task (T_1) Immediate recall	Task (T_2) Mental arithmetic	Task (T_3) Initial instructions
Experimental	1	1.07	1.02	1.02
	2	0.95	1.11	0.87
	3	0.94	0.93	0.89
	4	1.04	1.12	1.05
	5	0.96	1.01	0.96
	6	1.13	1.21	0.96
	7	0.90	0.89	1.14
	8	1.02	1.08	1.02
	9	1.04	1.16	0.93
	10	0.97	0.99	0.96
	Mean	1.00	1.05	0.98
Control	1	0.89	0.87	0.93
	2	0.96	0.97	0.88
	3	0.91	0.89	0.89
	4	1.03	0.96	0.97
	5	0.93	0.99	0.78
	6	0.87	0.89	1.00
	7	0.91	0.88	0.93
	8	0.98	0.89	0.86
	9	0.83	0.89	0.86
	10	0.80	0.87	0.90
	11	0.92	0.98	0.94
	Mean	0.91	0.92	0.91

TABLE II

Mean log ratios of the average EEG power during each task to the average EEG power during the resting period, i.e., log (PT/PR) * for each group (controls and experimentals) for each task.

Task	Controls (normal children)	Experimentals (LD children)
Immediate recall (T_1)	−0.096	0.000
Mental arithmetic (T_2)	−0.089	0.047
Initial instructions (T_3)	−0.101	−0.024

* PT, mean power of the alpha frequency during task work time; PR, mean power of the alpha frequency during the resting time.

T_3 condition than did the LD group. Using Fisher's Exact Test for a 2 by 2 contingency table, this difference is statistically significant at the $P < 0.01$ level (Beyer 1968). For the tasks (T_1 and T_2) and the initial instruction period (T_3) combined, the difference in alpha attenuation between LD and control groups was significant at the $P < 0.001$ level ($T^2 = 30.1$; $F = 9.99$; $df = 3, 17$).

To determine the statistical significance of

TABLE IV

Ratios of average EEG power during immediate recall task (T_1) to average EEG power during rest period showing individual subjects in order from least to most alpha attenuation.

Subject	Rank	Score	
		Experimental group	Control group
E6	1	1.13	
E1	2	1.07	
E4	3.5	1.04	
E9	3.5	1.04	
C4	5		1.03
E8	6	1.02	
C8	7		0.98
E10	8	0.97	
E5	9.5	0.96	
C2	9.5		0.96
E2	11	0.95	
E3	12	0.94	
C5	13		0.93
C11	14		0.92
C3	15.5		0.91
C7	15.5		0.91
E7	17	0.90	
C1	18		0.89
C6	19		0.87
C9	20		0.83
C10	21		0.80

TABLE III

Simultaneous confidence intervals for the control population mean minus the experimental population mean as shown in Table I.

Tasks	95% Interval	90% Interval	75% Interval
Immediate recall (T_1)	(−0.198, +0.006) width: 0.209 Includes 0, not significant	(−0.185, −0.007) width: 0.175 Excludes 0, significant	(−0.166, −0.026) width: 0.140 Excludes 0, significant
Mental arithmetic (T_2)	(−0.246, −0.026) width: 0.220 Excludes 0, significant	(−0.232, −0.040) width: 0.192 Excludes 0, significant	(−0.212, −0.060) width: 0.152 Excludes 0, significant
Initial Instructions (T_3)	(− 0.183, +0.029) width: 0.212 Includes 0, not significant	(−0.169, +0.019) width: 0.188 Includes 0, not dignificant	−0.149, −0.005) width: 0.144 Excludes 0, significant

the group differences in the three conditions separately, simultaneous confidence intervals were computed (Morrison 1967). This statistical method takes account of the non-independence of scores for each subject across the three conditions. Table III shows that the alpha attenuation difference between LD and normal control boys was significantly different at the $P < 0.05$ level for the mental arithmetic (T_2) conditions. The immediate recall (T_1) condition had a $P < 0.10$ but did not reach the 0.05 significance level. For the initial instruction condition (T_3), the difference was not significant at the 0.05 level. Tables IV, V, and VI demonstrate the considerable degree of both overlap and separation of scores of the two groups in the three conditions.

TABLE V

Ratios of average EEG power during mental arithmetic task (T_2) to the average EEG power during rest period showing individual subjects in order from least to most alpha attenuation.

Subject	Rank	Score	
		Experimental group	Control group
E6	1	1.21	
E9	2	1.16	
E4	3	1.12	
E2	4	1.11	
E8	5	1.08	
E1	6	1.02	
E5	7	1.01	
E10	8	0.99	
C5	8		0.99
C11	10		0.98
C2	11		0.97
C4	12		0.96
E3	13	0.93	
E7	14.5	0.89	
C3	14.5		0.89
C6	14.5		0.89
C8	14.5		0.89
C9	14.5		0.89
C7	19		0.88
C1	20.5		0.87
C10	20.5		0.87

TABLE VI

Ratios of average EEG power during the initial instruction period (T_3) to the average EEG power during the rest period showing the subjects in order from least to most alpha attenuation.

Subject	Rank	Score	
		Experimental group	Control group
E7	1	1.14	
E4	2	1.05	
E1	3.5	1.02	
E8	3.5	1.02	
C6	5		1.00
C4	6		0.97
E5	7.5	0.96	
E6	7.5	0.96	
E10	7.5	0.96	
C11	10		0.94
E9	11.5	0.93	
C1	11.5		0.93
C7	11.5		0.93
C10	14		0.90
E3	15.5	0.89	
C3	15.5		0.89
C2	17.5		0.88
C9	17.5		0.88
E2	19	0.87	
C8	20		0.86
C5	21		0.78

In addition to greater alpha attenuation, the normal control children also responded correctly to more of the problems. On the average the control group boys gave correct answers to 12 of the 16 problems; the LD boys averaged 7 correct responses.

Discussion

The results of this research show that the EEG alpha activity of normal boys attenuated more during experimental tasks chosen for this study than did that of the experimental group of LD boys. Table II shows that this

difference occurred in all conditions studied. Tables IV—VI show that although the group means differed in the expected direction in all three conditions, there was considerable overlap in the individual scores in conditions T_1 and T_3. Only on T_2 (mental arithmetic) were the groups consistently different, with only 2 of the LD boys having scores which overlapped with the control group boys (Table V).

These differences were statistically significant for the three conditions together, and for T_2 separately, but not for T_1 or T_3 considered separately. Since the groups contained only 10 and 11 subjects respectively, additional data from further experimentation are needed to determine whether or not the observed trends would be confirmed if the number of subjects was larger.

EEG research has shown that alpha attenuation is a concomitant of attention and mental effort (Glass 1959, 1960, 1964); therefore, the results of this experiment are consistent with the possibility of attention deficits in these LD children. Such deficits could result from a number of causes. It is possible that: (1) specific neurological dysfunction in LD children interferes with the focusing of attention, and therefore there is less alpha attenuation; (2) some LD children may be slow to mature in CNS function, their maturational lag accounting for both an attention deficit and lesser alpha attenuation; (3) the observed differences are of psychological, rather than neurological, origin; or (4) combinations of the above three possibilities. Further research is needed in order to determine the relative importance of each of these possibilities; more precise understanding of cognitive functioning will help us to understand these relationships.

Research in this area has been hampered by inconsistent criteria used in diagnosing learning disabilities. In these children there is often no observable brain damage or trauma, a clearly defined medical syndrome has not been established, and almost all definitions to date have been totally behavioral (Sroufe 1974). Furthermore, "traditional" approaches to EEG may not be particularly rewarding in LD children and a variety of findings have been observed (Daveau 1959; Capute et al. 1968; Satterfield 1972). Previous studies of the relationship between abilities and EEG "have been hampered by a rigid adherence to traditional methodological approaches" (Vogel et al. 1968), and similar criticisms have been stated by Mundy-Castle (1957) and Vogel and Broverman (1964, 1966). Vogel and Broverman (1964) argued that EEG measurements taken during an active intellectual effort would most probably show the greatest relationship to cognitive measurements. Furthermore, Brazier (1968) has stated that "the development from the concept of the electroencephalogram as primarily an aid in assessing the normality of the resting 'inactive' brain to a dynamic formulation of it as an indicator of the brain in action, has, with its emphasis on function rather than on morphology, suggested a wider application of the EEG to the study of behavior . . .". Knott (1938) came to this same realization in the early years of EEG recording from the human subject: "While the brain potential (electroencephalographic) technique undoubtedly holds the key to many psychological problems in the field of cerebration, the great majority of the published experimental work has been done under conditions of minimal stimulation rather than under psychologically 'normal' conditions . . .".

Thus, it is essential to explore disturbances of attention in LD children in relationship to EEG parameters while they are actively performing tasks. Furthermore, the meaning of the results may be enhanced when stimuli analagous to those that are difficult in the school learning environment are used. A search for anomalous responses to stimuli might be expected to yield better results than the search for abnormalities in the resting EEG. Therefore, the purpose of this study was to begin exploration of an objective physiological concomitant of attention in LD and control children.

Summary

The parieto-occipital EEG and power spectra were recorded during resting intervals, while listening to instructions on tape, and during active performance of mental arithmetic and immediate recall to determine if 10 boys with learning disabilities (LD) would show less alpha attenuation than 11 normal controls. The LD boys showed less alpha attenuation than the normal control boys. In addition, the control boys responded correctly to more of the problems. Since attention deficits are found in LD children, and attenuation of the parieto-occipital alpha rhythm is correlated with attention, these results show that a deficit of the latter can be associated with disturbances in the former.

Résumé

Atténuation de l'alpha estimée par ordinateur au cours de résolution de problèmes chez l'enfant avec troubles cognitifs

L'EEG pariéto-occipital et les spectres de puissance ont été enregistrés au cours d'intervalles de repos, lors de l'écoute d'instructions sur magnétophone, et au cours de la réalisation active de calcul mental et de souvenir immédiat afin de déterminer si 10 garçons présentant des troubles cognitifs montreraient moins d'atténuation d'alpha que 11 contrôles normaux. Les garçons avec troubles cognitifs ont effectivement montré moins d'atténuation d'alpha que les garçons normaux de contrôle. Enoutre, les sujets de contrôle répondent correctement à plus de problèmes. Puisqu'on observe des troubles de l'attention chez les enfants avec troubles cognitifs, et que l'atténuation de rythmes alpha parieto-occipital est corrélée à l'attention, ces résultats montrent qu'un déficit de celle-ci peut s'associer à des perturbations de celle-là.

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